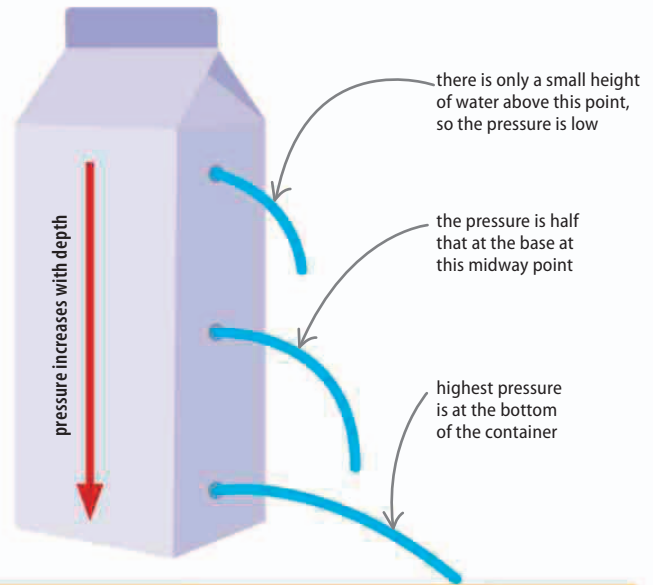


Water pressure

Water is far denser than air. This means that as one goes deeper under water the pressure increases rapidly. On Earth, the water pressure at 10 m (33 ft) depth is about one “atmosphere,” which is roughly the air pressure at sea level. At a depth of 20 m (66 ft), the water pressure is about two atmospheres, and at 30 m (99 ft) it is about three atmospheres, and so on.

▷ Under pressure

In this milk carton, the pressure increases toward the base, so water will squirt out under greater pressure from the bottom hole than from the top one.

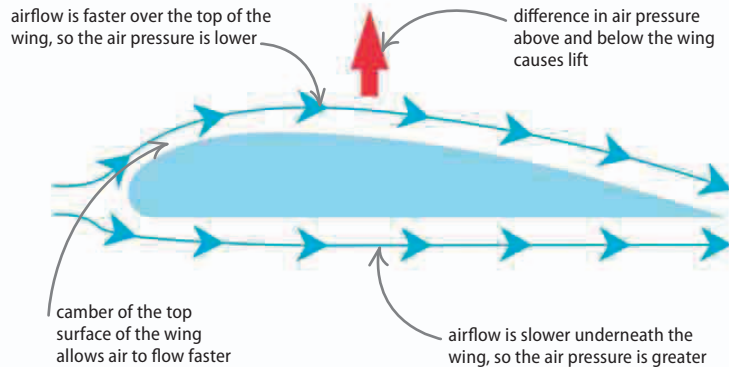


Bernoulli effect

Pressure varies according to the motion of a medium—this is called the Bernoulli effect. In an airplane wing, the top surface of the wing has more camber (longer curve) than the bottom surface, so the air flows faster over the top of the wing than it does underneath.

▷ Taking flight

There is less air pressure above the wing than there is beneath the wing. The difference in the air pressure above and below the wing causes lift.

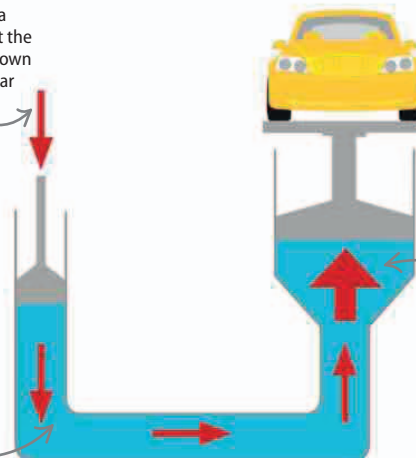


Hydraulics

In an hydraulic system, a liquid (often oil) is used to transfer force from one place to another. Usually, hydraulic systems also convert a low force at one place to a higher one at another. Hydraulic systems rely on the fact that liquids (unlike gases) are almost incompressible—if they are pressed, rather than reducing in volume they force objects like pistons to move away.

narrow pipe means only a small force is needed, but the piston must be pushed down a long way to move the car up a short distance

pressure throughout the oil is the same



◁ Hydraulic multiplication

The increase in force created by an hydraulic system can be calculated from the areas of the two sides (shown here in cross-section): if one side has twice the area of the other, the force will be doubled.